FIELD INTELLIGENCE SYSTEM

PHANTOM

The field intelligence tool built for the rack, not the desk.

Proof of Concept · Public Sources · Needs Field Validation

~16,300	70	6	14	100%
LINES OF CODE	DEPLOY FUNCTIONS	TABS	GPU PLATFORMS	OFFLINE CAPABLE

John Hamilton

Data Center Technician · CoreWeave AUS-01

linkedin.com/in/johnhamiltonleadtech

THE PROBLEM

The knowledge lives in someone's head.

When they leave the shift, it goes with them.

A CoreWeave DCT troubleshooting a down link on shift shouldn't be searching Confluence. The reference is buried in PDFs, wikis, and Slack threads. Most tools need a network connection that may not exist on the floor. Every fix that resolves an incident disappears the moment the shift ends — taking hard-won knowledge with it.

THE PITCH

PHANTOM puts the answer in your pocket.

One file. Works offline. Learns from your own experience on your own racks. When your hands are full and something's wrong, you ask it out loud and get an answer back. It knows your stack. It remembers your fixes. And the more you use it, the smarter it gets — without a database, without a backend, without IT.

IMPORTANT NOTE

PHANTOM is a proof of concept built entirely from public vendor documentation. The app is concept-approved — built, functional, and live on GitHub — but not yet data-approved. It runs on general knowledge, not real AUS-01 site data. Until permission is granted to feed it actual rack configs, optic inventory, and port maps, it operates as a smart reference tool. Full site data integration is the natural next step once approved.

HOW IT ALL CONNECTS

Every feature in PHANTOM is designed to feed every other feature. The triage flowchart generates structured fault data. That data feeds Ghost Echo. Ghost Echo feeds the hypothesis engine. The hypothesis engine gets smarter with every logged fix. The Build Lead deployment engine tracks every rack from EDP ingest through validation sign-off. See page 4 for the full breakdown.

MAJOR BENEFITS

Never lose a fix again.

Ghost Echo — Institutional Memory

Captures every resolved incident — what you did and whether it worked — and surfaces it the next time the same fault appears on the same rack, on any shift, by any employee.

Phantom Hypothesis — Ranked Actions

Turns your logged fix history into ranked hypotheses. Describe the fault and get a confidence-scored action plan built from real field data — not a generic checklist.

Voice Assistant — Hands-Free Answers

Hold-to-talk voice assistant with full knowledge of your infrastructure stack. Ask about optic thresholds, CLI commands, or fault codes out loud. Hands-free, eyes-free, on the floor.

Rack Map — Visual Elevation Builder

Paste any elevation format — spreadsheet, plain text, Visio, anything. PHANTOM renders a visual 42U rack instantly, color-coded by device type. Tap any device for a full detail sheet.

Reference Library — Offline Everything

Fiber types, optic specs, CLI commands, PDU diagrams, GPU fault codes — all embedded. No signal needed. No Confluence. Organized the way a shift works.

Build Lead — Deployment Command Engine

Paste an EDP, get instant local parsing of rack manifests, fiber connections, and optic counts. 5-phase pipeline per rack (MECH/PWR/NET/COMP/VAL) with tappable task checklists, tech assignments, phase gates, blocker tracking, and AI-powered EDP validation. Track an entire NVL72 pod buildout from your phone.

THE INTEGRATED SYSTEM

Every feature feeds every other feature.

PHANTOM is not a collection of separate tools bolted together. Every feature is architected to generate data that another feature consumes. The result is a system that gets measurably smarter every time a tech completes a shift.

Triage > Ghost Echo

The 7-step triage flowchart structures fault type, rack, platform, and actions taken. That data becomes a Ghost Echo record the moment the fix is confirmed.

Ghost Echo > Hypothesis Engine

Every logged fix is scored by the hypothesis engine. The more fixes logged on a fault type, the higher the confidence score. Ghost Echo is the training data.

CLI Parser > Ghost Echo

Parsed fault fingerprints — Rx power, error counters, port state — attach to the Ghost Echo record automatically.

Rack Map > Optic Selector

The rack map knows what platform is in each U position. Tap a device, hit optic selector — platform pre-fills automatically.

Port Map Validator > Triage

A validated port map becomes the source of truth for triage. The tech sees exactly which ports are mapped and expected state.

Voice Assistant > Everything

Access to the entire knowledge base — optic specs, CLI commands, Ghost Echo history, hypothesis scores, SOP library. One question, every data source.

SOP Manager > Triage

Custom SOPs from senior techs feed into the triage flowchart as site-specific steps. The app learns your facility's procedures.

EDP Parser > Build Lead

A parsed EDP seeds rack records, device inventories, fiber connections, and optic counts directly into the deployment pipeline. Local parser — zero API calls.

Build Lead > Task Engine

Each phase generates a task checklist from templates. Tasks track completion per rack per phase. Phase auto-completes when all tasks are done.

FEATURES AT A GLANCE

Triage Flowchart
7-step down-link protocol. Tap each step as you progress.

CLI Reference
Arista, NX-OS, MLNX, InfiniBand, OOB — all offline, searchable.

Ghost Echo
Log every fix. Pattern-match your history across racks and shifts.

Port Map Validator
Paste any format. Clean, sorted, validated with flagged errors.

Voice Assistant
Ask any question out loud. Answer read back hands-free.

QR / Barcode Scanner
3-tier fallback. Scan asset tags, optics, rack labels.

Backup / Restore
Full export/import of all app data. Device migration ready.

Audit Trail
Append-only event log. Full traceability EDP ingest to sign-off.

Command Center
Multi-deployment dashboard. Phase heat map, aggregate stats.

Optic Selector
Pick a platform, get the exact optic spec for your distance and fiber.

PDU Load Balancer
Tap any PSU or phase to see the full circuit path highlighted.

Phantom Hypothesis
Fault + Rx power + history = ranked actions with confidence scores.

SOP Manager
Create, save, filter SOPs on device. Import from iOS Files.

Smart Search
Type anything — routes to the right tool or reference tab.

Shift Handoff
Structured end-of-shift handoff with auto-generated context.

Build Lead
70-function deployment engine. EDP ingest, 5-phase pipeline, optic ledger.

RBAC / Lead Mode
PIN-gated build lead access. SHA-256 hashed. Per-deployment.

Completion Reports
Auto-generated HTML/JSON reports from all deployment data.

THE BIGGER PICTURE

Built for CoreWeave employees. Phase 1 — internal only.

PHANTOM is designed first and foremost for CoreWeave's own Data Center Technicians. Every employee on the floor deals with the same problem — critical knowledge scattered across Confluence, Slack, and people's heads. There's no standardized way to ensure a tech follows the right triage sequence, documents their fix correctly, or escalates at the right threshold. It depends on who showed up that day.

PHANTOM changes that by becoming the live standard operating procedure in every employee's hand on the floor.

HOW IT HELPS THE TEAM

Standardize the Process

The 7-step triage flowchart becomes the consistent sequence every tech follows. Ghost Echo becomes the required incident log for every resolved fault. Consistent documentation across every shift.

Turn Field Data into Institutional Knowledge

Ghost Echo data aggregated across all employees reveals which fault types are being mishandled and which racks generate repeat incidents. Knowledge stays in the organization — not in someone's head.

New Employees Inherit Team Knowledge on Day One

As more techs log fixes, the Phantom hypothesis engine gets smarter. A new employee on their first shift inherits the logged experience of everyone who worked that rack before them — automatically.

THE VISION

CoreWeave's Data Center Technicians are some of the most technically skilled people in the industry — doing high-stakes work with tools that were never designed for them. PHANTOM is a step toward fixing that. It needs real testing against real workflows and real data. But the foundation is solid, the problem is real, and the potential is significant.

SECURITY ARCHITECTURE

How PHANTOM protects data at every layer.

PHANTOM is designed with a zero-trust approach to secrets and data. No API keys live in the app. No sensitive data leaves the device without explicit user action. Access to AI-powered features is gated behind authentication, and IT retains full control over provisioning and revocation.

CLOUDFLARE WORKER PROXY

All external API calls route through a Cloudflare Worker — a lightweight edge function that holds secrets in encrypted environment variables. The app never sees or stores any key. The Worker enforces origin validation, accepting requests only from the exact GitHub Pages domain.

- API keys stored in Cloudflare encrypted environment variables
- Origin-locked — rejects requests from any unauthorized domain
- Handles both Anthropic and Jira API tokens in one secure layer

WORK PROFILE SYSTEM

Sensitive features — AI assistant, Jira integration, Ghost Echo sync — are gated behind a company-issued work profile token. All user data is stored under the token's namespace in localStorage. IT controls issuance, expiration, and remote revocation.

- Token-gated access — no token, no sensitive features
- Data namespaced per token — isolated between users
- Three wipe triggers: manual logout, auto-expiry, IT revocation

DATA HANDLING

PHANTOM processes data on-device by default. AI queries are sent to the Anthropic API via the Cloudflare proxy, processed in memory, and returned — Anthropic does not store or train on API traffic. Locally saved data (Ghost Echo logs, SOPs, rack maps) persists only on the device and wipes automatically when the work profile is revoked.

- No persistent server-side storage of user queries or responses
- Local data tied to work profile — revoke token, wipe data
- Full audit trail of every AI query and Jira ticket accessed
- Client-side EDP parsing — zero API calls for structured CSV data
- PIN-gated RBAC — SHA-256 hashed, per-deployment, sessionStorage scoped
- Auto-retry on 429/529 with 8s/15s backoff and toast notification

ADD-ON: JIRA TICKET INTELLIGENCE

Walk to the rack already knowing everything.

A DCT gets assigned a Jira ticket. Instead of reading through paragraphs of notes and back-and-forth comments — they open PHANTOM, enter the ticket number, and get back a complete mission brief in seconds.

```
PHANTOM MISSION BRIEF  ·  CW-4821  READY

  RACK              R-07  ·  Row 3, Position 14
  NODE              H100 SXM5  ·  GPU-04
  FAULT             Rx power below threshold  ·  Eth1/12  ·  -18 dBm
  SUSPECTED         Dirty ferrule or degraded QSFP28
  OPTIC NEEDED      QSFP28-SR4  ·  confirm rack elevation first
  TOOLS             Fiber scope  ·  IEC cleaning kit  ·  spare QSFP28
  FIRST ACTION      Clean both ferrule end-faces before swapping optic
  ESCALATE IF       Rx does not recover above -14 dBm after cleaning
```

WHY THIS MATTERS

No More Wasted Rack Walks

A tech reads a ticket, goes to the rack, realizes they forgot a tool, walks back. 20 minutes wasted per incident. PHANTOM eliminates that loop.

Team Knowledge Becomes Structured

Senior tech notes buried in Jira comments get read, understood, and surfaced automatically in the brief.

Every Employee Gets the Same Quality Brief

A tech on their first week gets the same mission brief as someone with 3 years on the floor.

Ghost Echo Closes the Loop

After the fix is logged, Ghost Echo updates automatically. Ticket closed, field intelligence updated.

JIRA INTEGRATION — TECHNICAL

Architecture and security details.

DATA FLOW

PHANTOM APP → CLOUDFLARE PROXY → JIRA REST API → RAW TICKET DATA

CLOUDFLARE PROXY → ANTHROPIC API → MISSION BRIEF → PHANTOM APP

SECURITY ARCHITECTURE

Jira token fully secured.
Lives in Cloudflare's encrypted environment alongside the Anthropic key. Never in the app code, never on GitHub.

Work profile gated.
Jira integration only activates for employees with a valid company-issued token. IT controls who gets access.

Data stays on device.
Mission briefs generated on demand. Nothing stored permanently unless the employee explicitly saves it.

Remote wipe covered.
If a device is lost, IT revokes the work profile token. All saved data wipes on next app open.

Full audit trail.
Every ticket accessed through PHANTOM is logged with timestamp and employee profile ID.

THE ONE SENTENCE PITCH

"A CoreWeave tech opens their Jira ticket in PHANTOM and walks to the rack already knowing exactly what's wrong, what to bring, and what to do first."

SYS.STATUS: READY // BUILD INTEGRITY: VERIFIED // FIELD VALIDATION: PENDING

PHANTOM

FIELD INTELLIGENCE SYSTEM · PROOF OF CONCEPT · INTERNAL USE ONLY

John Hamilton

Data Center Technician · CoreWeave AUS-01

linkedin.com/in/johnhamiltonleadtech

darkmatter024.github.io/phantom/dct-ios.html

phantom@aus-01 ~ /sys/manifest

```
> phantom --version
  PHANTOM v2.0-rc  //  Field Intelligence System

> phantom --manifest
  CODEBASE       ~16,300 lines  //  single-file PWA
  ARCHITECTURE   zero-dependency HTML  //  no build step
  AI ENGINE      Claude API via Cloudflare proxy
  EDP PARSER     client-side deterministic  //  zero API
  DEPLOY ENGINE  70 functions  //  5-phase pipeline
  TASK ENGINE    23 templates  //  auto-complete  //  assignments
  RBAC           PIN-gated lead mode  //  SHA-256 hashed
  CONNECTIVITY   100% offline core  //  AI requires network
  PLATFORMS      14 GPU architectures supported
  DATA LAYER     21 localStorage keys  //  backup/restore
  SECURITY       origin-locked proxy  //  token-gated access

> phantom --status
  CORE FEATURES  [=====================================] READY
  DEPLOY ENGINE  [=====================================] READY
  TASK ENGINE    [=====================================] READY
  AI FEATURES    [=====================================] GATED
  SITE PROFILE   [////////////////////                 ] PENDING
  FIELD TESTED   [                                     ] AWAITING
```